

Head First Design Patterns



Wouldn't it be dreamy if
there was a Design Patterns
book that was more fun than going
to the dentist, and more revealing
than an IRS form? It's probably
just a fantasy...

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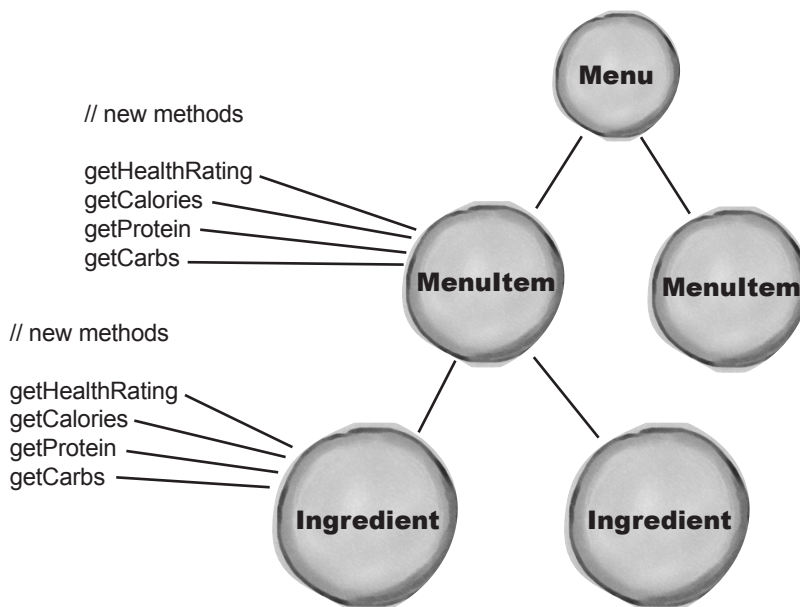
Visitor

Use the Visitor Pattern when you want to add capabilities to a composite of objects and encapsulation is not important.

A scenario

Customers who frequent the Objectville Diner and Objectville Pancake House have recently become more health conscious. They are asking for nutritional information before ordering their meals. Because both establishments are so willing to create special orders, some customers are even asking for nutritional information on a per ingredient basis.

Lou's proposed solution:

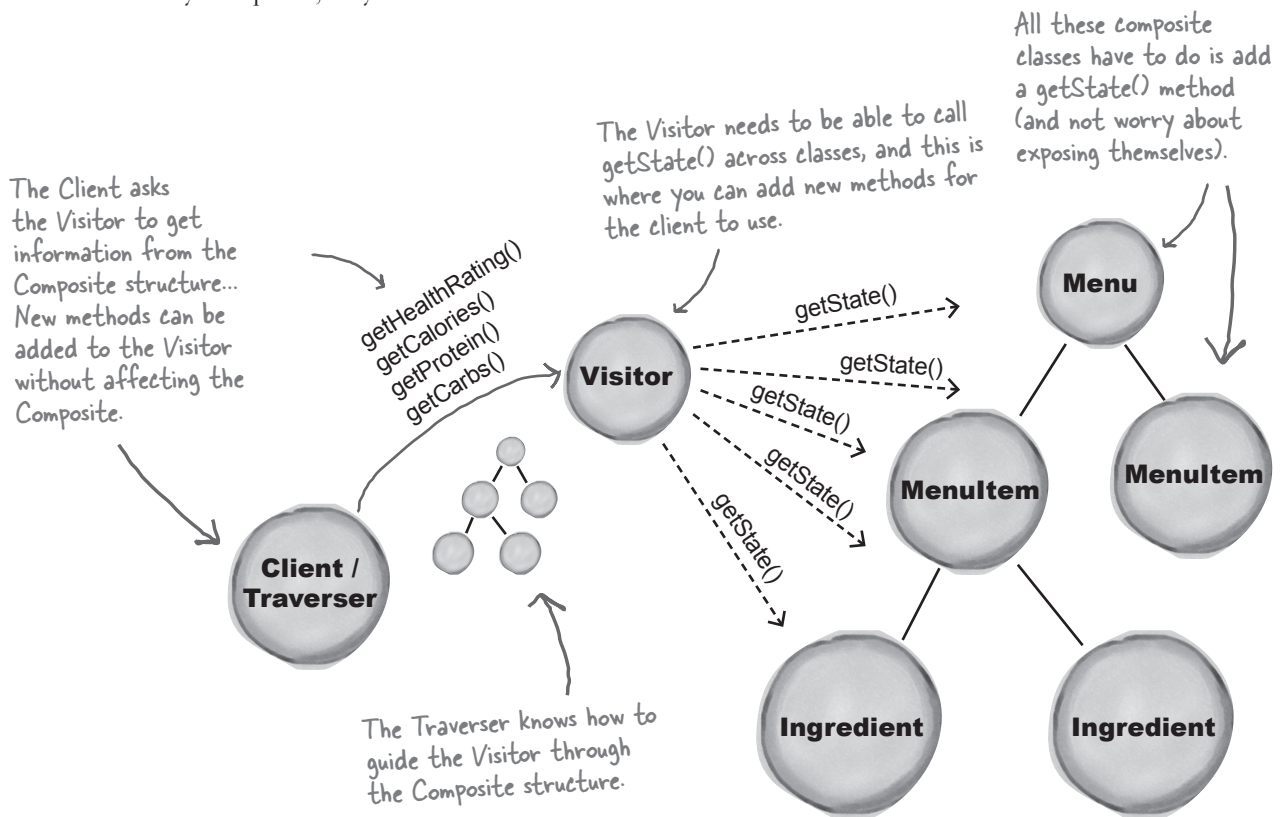


Mel's concerns...

“Boy, it seems like we’re opening Pandora’s box. Who knows what new method we’re going to have to add next, and every time we add a new method we have to do it in two places. Plus, what if we want to enhance the base application with, say, a recipes class? Then we’ll have to make these changes in three different places...”

The Visitor drops by

The Visitor works hand in hand with a Traverser. The Traverser knows how to navigate to all of the objects in a Composite. The Traverser guides the Visitor through the Composite so that the Visitor can collect state as it goes. Once state has been gathered, the Client can have the Visitor perform various operations on the state. When new functionality is required, only the Visitor must be enhanced.



Visitor Benefits

- Allows you to add operations to a Composite structure without changing the structure itself.
- Adding new operations is relatively easy.
- The code for operations performed by the Visitor is centralized.

Visitor Drawbacks

- The Composite classes' encapsulation is broken when the Visitor is used.
- Because the traversal function is involved, changes to the Composite structure are more difficult.